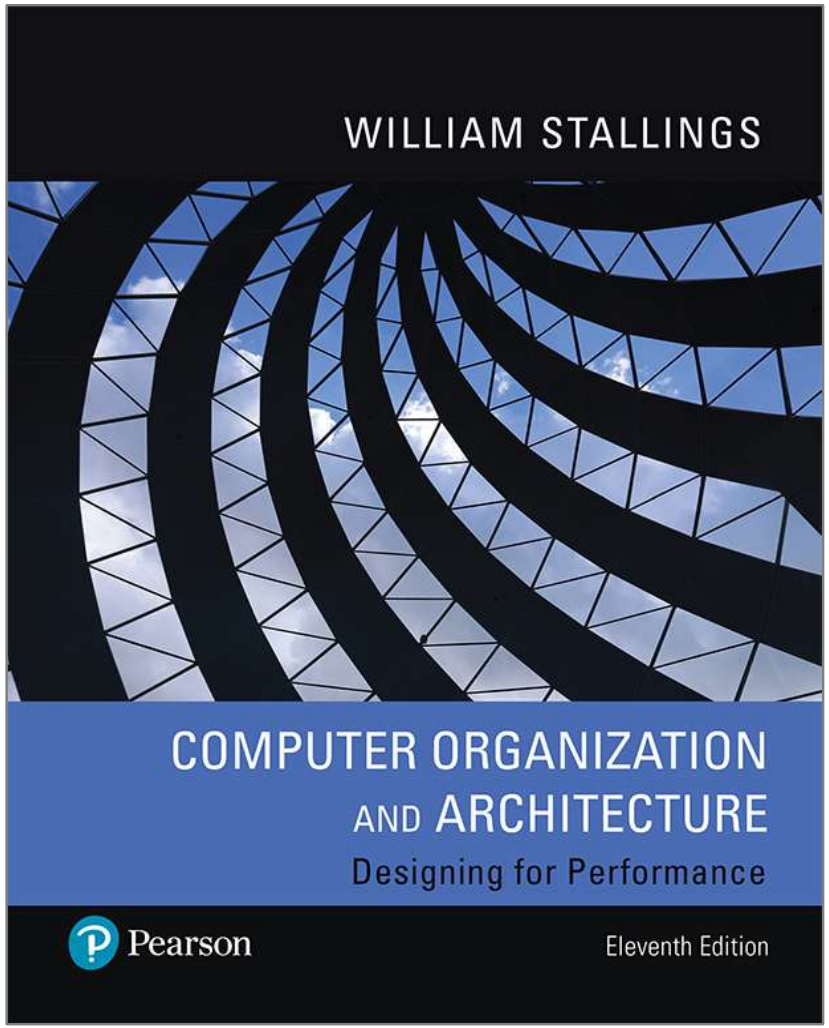
Computer Organization and Architecture

Designing for Performance

11th Edition



Chapter 2

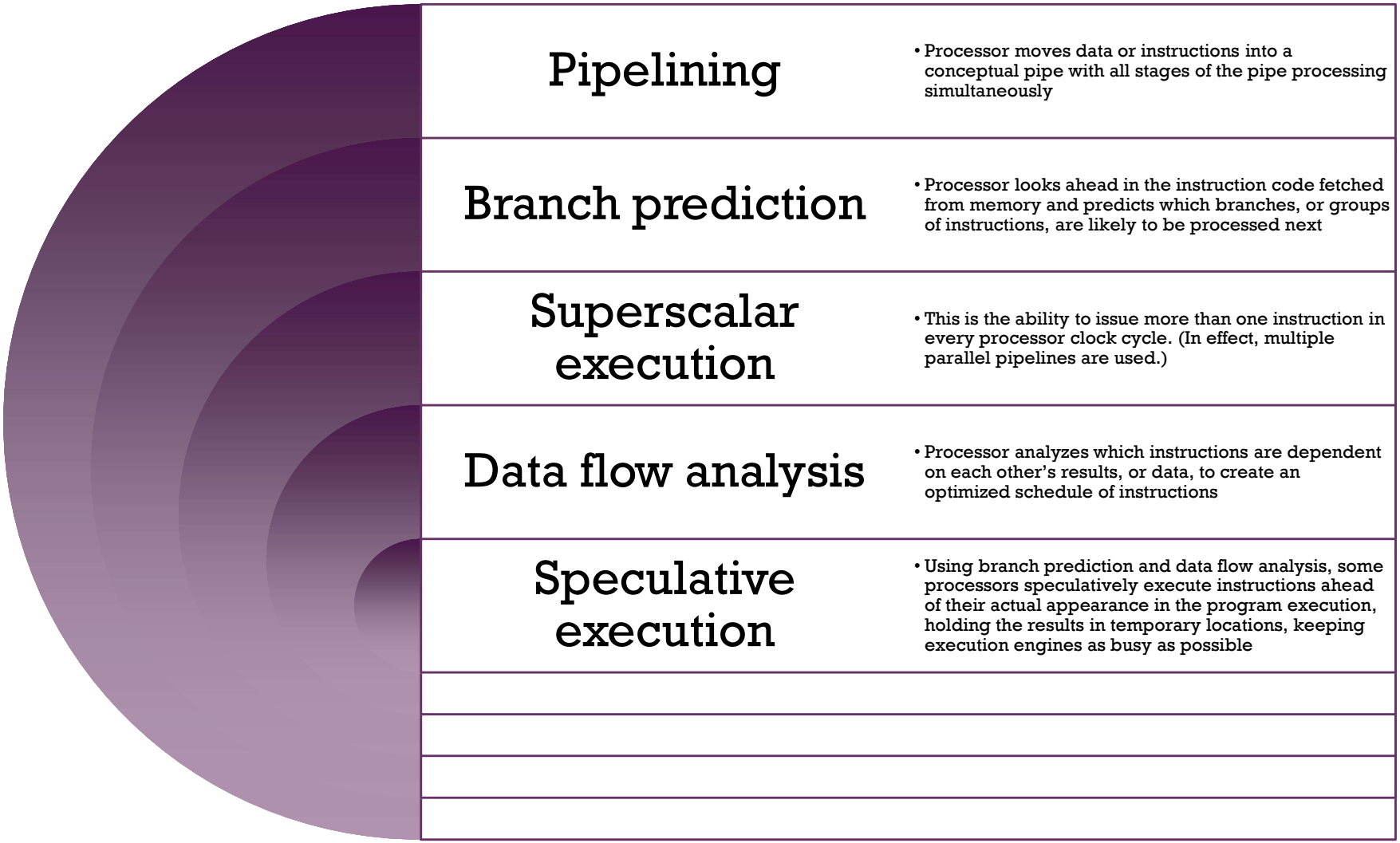
Performance Concepts

Designing for Performance

- The cost of computer systems continues to drop dramatically, while the performance and capacity of those systems continue to rise equally dramatically
- Today's laptops have the computing power of an IBM mainframe from 10 or 15 years ago
- Processors are so inexpensive that we now have microprocessors we throw away
- Desktop applications that require the great power of today's microprocessor-based systems include:
  - Image processing
  - Three-dimensional rendering
  - Speech recognition
  - Videoconferencing
  - Multimedia authoring
  - Voice and video annotation of files
  - Simulation modeling
- Businesses are relying on increasingly powerful servers to handle transaction and database processing and to support massive client/server networks that have replaced the huge mainframe computer centers of yesteryear
- Cloud service providers use massive high-performance banks of servers to satisfy high-volume, high-transaction-rate applications for a broad spectrum of clients

Microprocessor Speed

Techniques built into contemporary processors include:



Pipelining	• Processor moves data or instructions into a conceptual pipe with all stages of the pipe processing simultaneously
Branch prediction	• Processor looks ahead in the instruction code fetched from memory and predicts which branches, or groups of instructions, are likely to be processed next
Superscalar execution	• This is the ability to issue more than one instruction in every processor clock cycle. (In effect, multiple parallel pipelines are used.)
Data flow analysis	• Processor analyzes which instructions are dependent on each other's results, or data, to create an optimized schedule of instructions
Speculative execution	• Using branch prediction and data flow analysis, some processors speculatively execute instructions ahead of their actual appearance in the program execution, holding the results in temporary locations, keeping execution engines as busy as possible

Performance Balance

- Adjust the organization and architecture to compensate for the mismatch among the capabilities of the various components
- Architectural examples include:

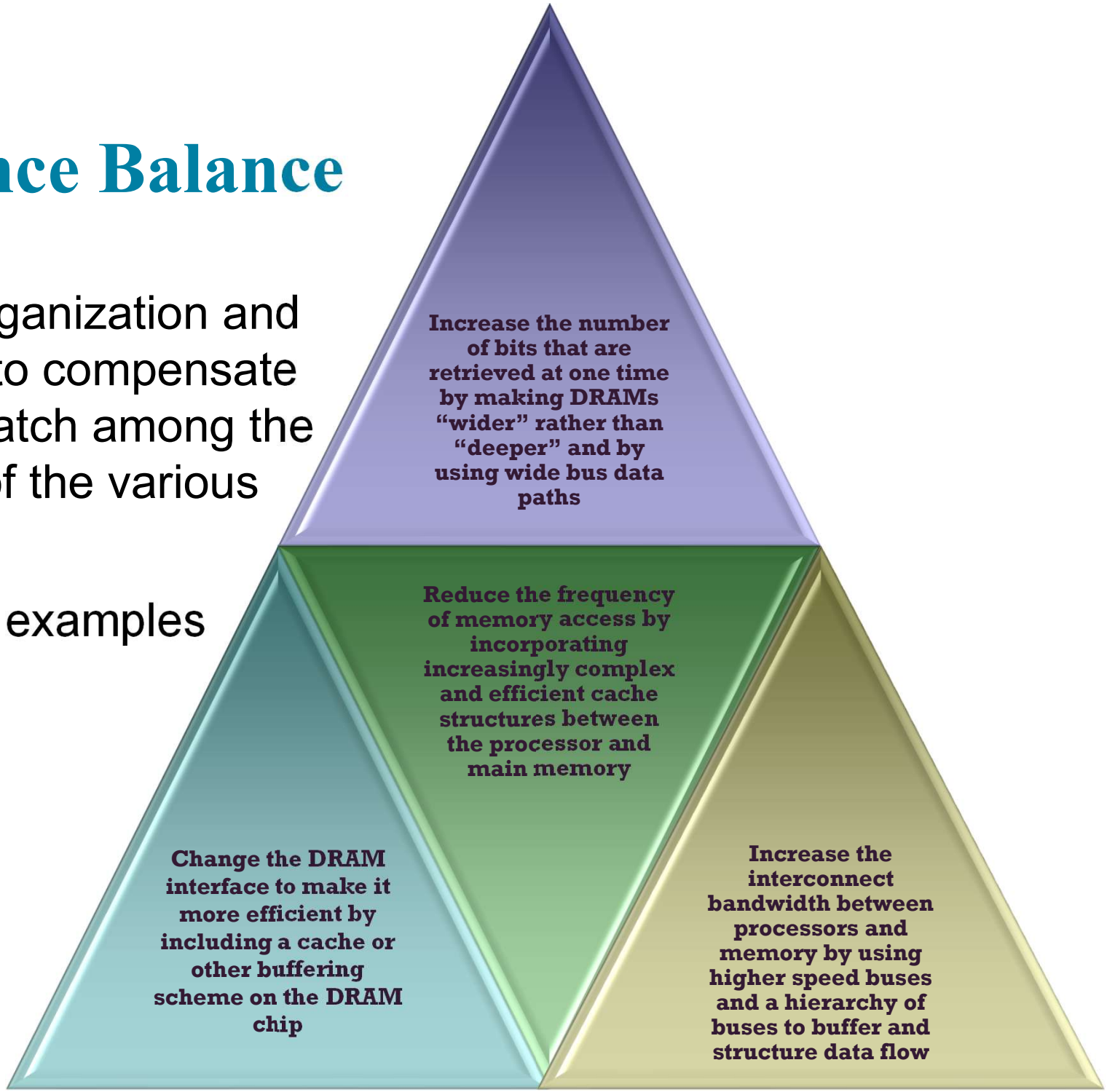


Figure 2.1

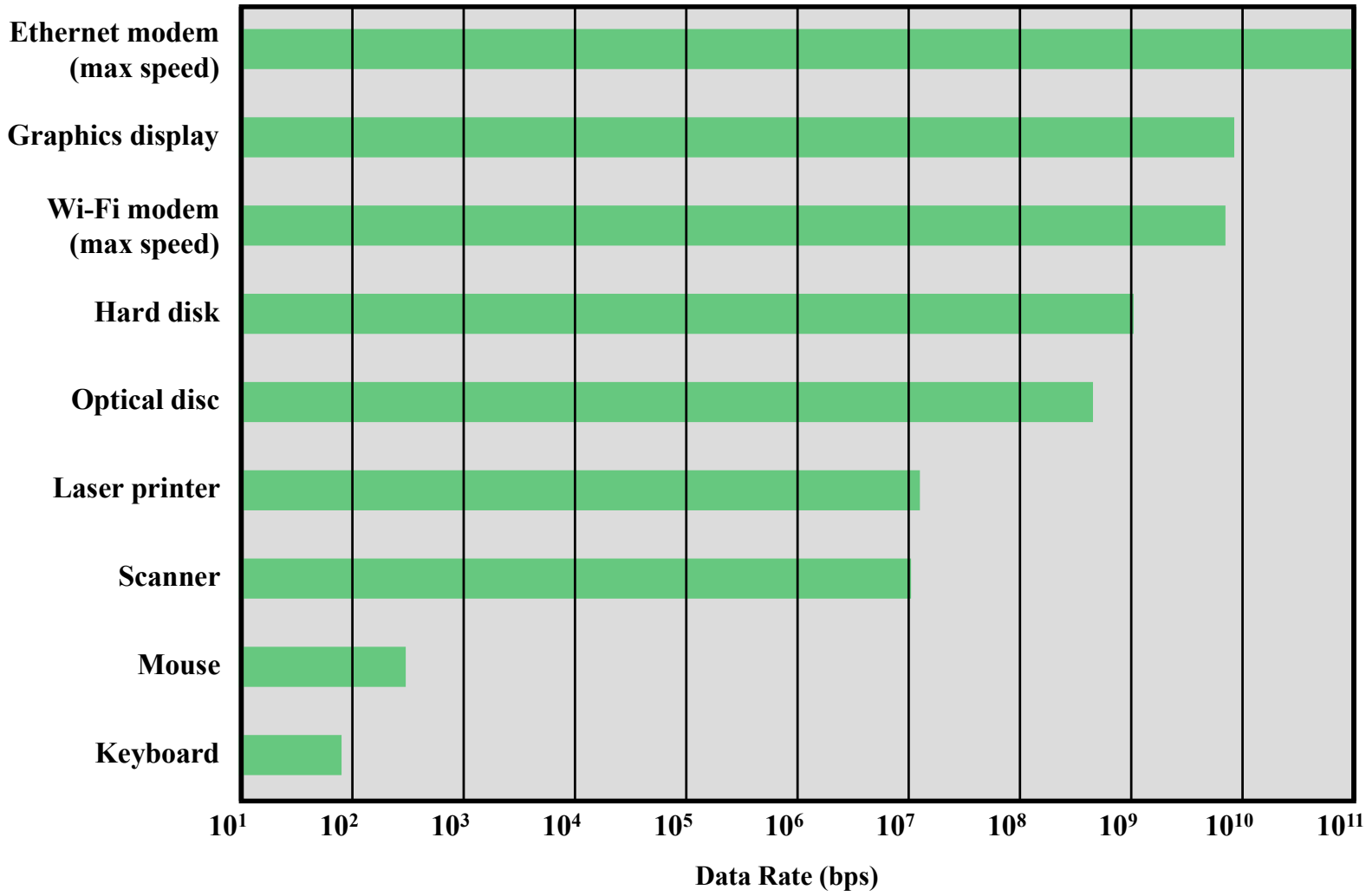


Figure 2.1 Typical I/O Device Data Rates

Improvements in Chip Organization and Architecture

- Increase hardware speed of processor
  - Fundamentally due to shrinking logic gate size
    - More gates, packed more tightly, increasing clock rate
    - Propagation time for signals reduced
- Increase size and speed of caches
  - Dedicating part of processor chip
    - Cache access times drop significantly
- Change processor organization and architecture
  - Increase effective speed of instruction execution
  - Parallelism

Problems with Clock Speed and Logic Density

- Power
  - Power density increases with density of logic and clock speed
  - Dissipating heat
- RC delay
  - Speed at which electrons flow limited by resistance and capacitance of metal wires connecting them
  - Delay increases as the RC product increases
  - As components on the chip decrease in size, the wire interconnects become thinner, increasing resistance
  - Also, the wires are closer together, increasing capacitance
- Memory latency and throughput
  - Memory access speed (latency) and transfer speed (throughput) lag processor speeds

Figure 2.2

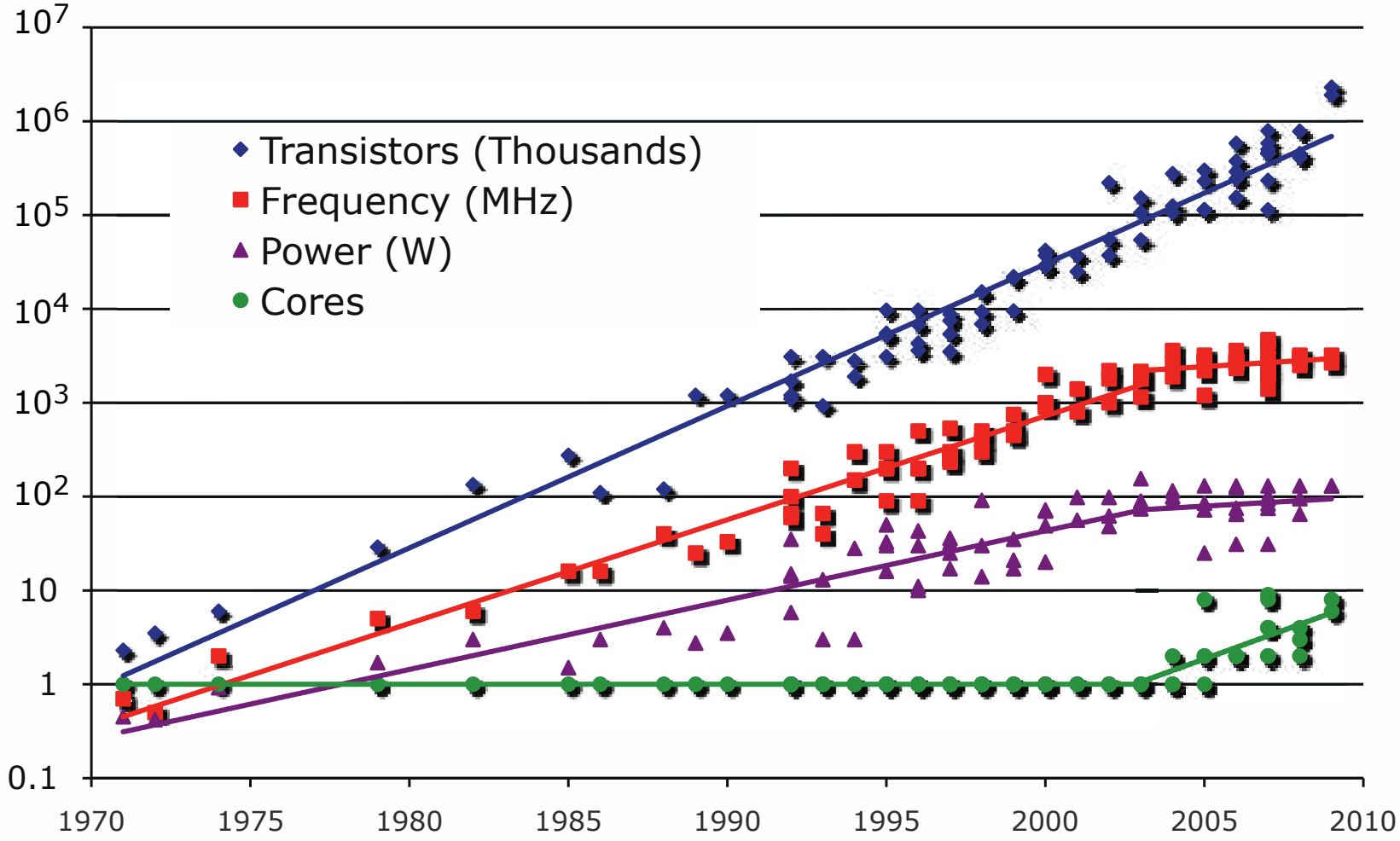
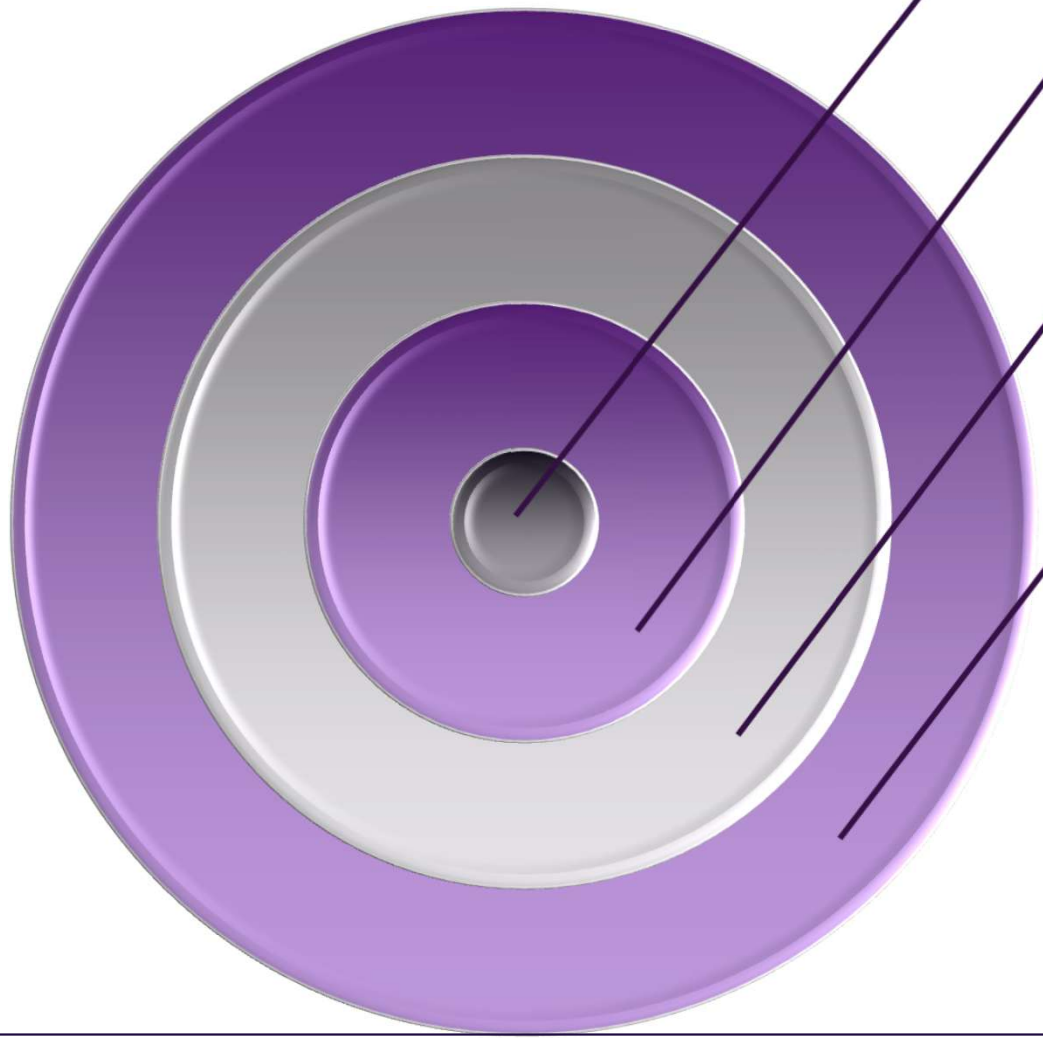


Figure 2.2 Processor Trends

Multicore



The use of multiple processors on the same chip provides the potential to increase performance without increasing the clock rate

Strategy is to use two simpler processors on the chip rather than one more complex processor

With two processors larger caches are justified

As caches became larger it made performance sense to create two and then three levels of cache on a chip

Many Integrated Core (MIC) Graphics Processing Unit (GPU)

MIC

- Leap in performance as well as the challenges in developing software to exploit such a large number of cores
- The multicore and MIC strategy involves a homogeneous collection of general purpose processors on a single chip

GPU

- Core designed to perform parallel operations on graphics data
- Traditionally found on a plug-in graphics card, it is used to encode and render 2D and 3D graphics as well as process video
- Used as vector processors for a variety of applications that require repetitive computations

Figure 2.5

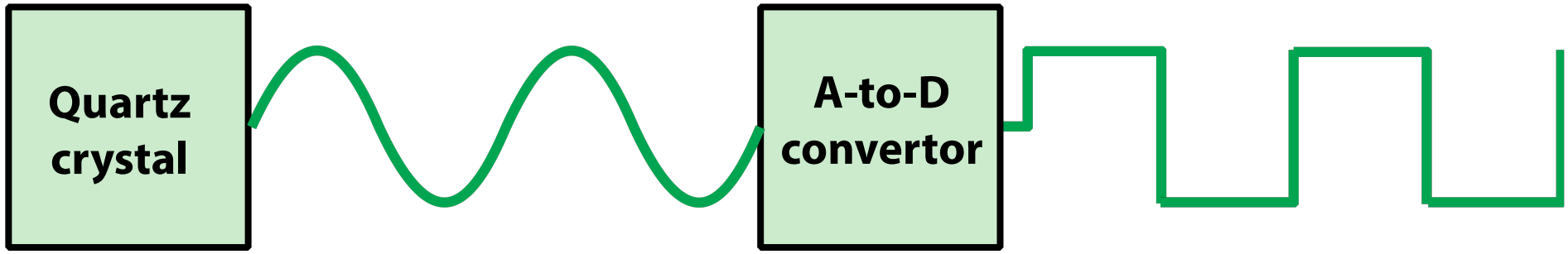
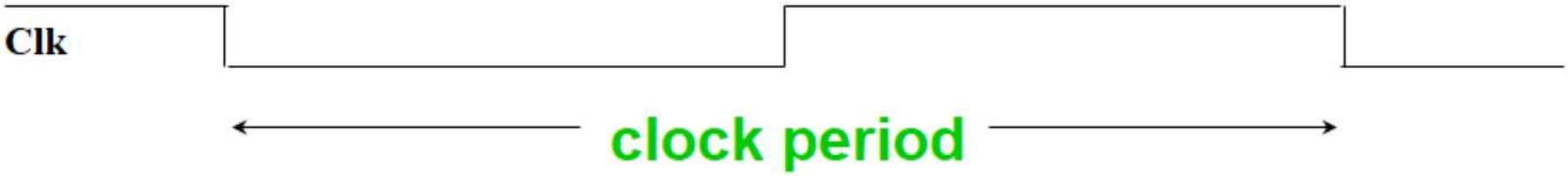


Figure 2.5 System Clock

Computer Clocks

- A computer clock runs at a constant rate and determines when events take placed in hardware.



- The clock cycle time is the amount of time for one clock period to elapse (e.g. 5 ns).
- The clock rate is the inverse of the clock cycle time.
- For example, if a computer has a clock cycle time of 5 ns, the clock rate is:

$$\frac{1}{5 \times 10^{-9}\,\text{sec}} = 200\ \text{MHz}$$

Table 2.1 Performance Factors and System Attributes

	Ic	*p*	*m*	*k*	τ
Instruction set architecture	X	X			
Compiler technology	X	X	X		
Processor implementation		X			X
Cache and memory hierarchy				X	X

Computing CPU time

- The time to execute a given program can be computed as

  CPU time = CPU clock cycles x clock cycle time

  Since clock cycle time and clock rate are reciprocals

  CPU time = CPU clock cycles / clock rate

- The number of CPU clock cycles can be determined by

  CPU clock cycles = (instructions/program) x (clock cycles/instruction)

  = Instruction count x CPI

  which gives

  CPU time = Instruction count x CPI x clock cycle time

  CPU time = Instruction count x CPI / clock rate

- The units for this are

$$\text{seconds} = \frac{\text{instructions}}{\text{program}} \times \frac{\text{clock cycles}}{\text{instruction}} \times \frac{\text{seconds}}{\text{clock cycle}}$$

Example of Computing CPU time

- If a computer has a clock rate of 50 MHz, how long does it take to execute a program with 1,000 instructions, if the CPI for the program is 3.5?

- Using the equation

  CPU time = Instruction count x CPI / clock rate

  gives

  CPU time = $1000 \times 3.5 / (50 \times 10^6)$

- If a computer's clock rate increases from 200 MHz to 250 MHz and the other factors remain the same, how many times faster will the computer be?

$$\frac{\text{CPU time old}}{\text{CPU time new}} = \frac{\text{clock rate new}}{\text{clock rate old}} = \frac{250\ \text{MHz}}{200\ \text{MHZ}} = 1.25$$

- What simplifying assumptions did we make?

Computing CPI

- The CPI is the average number of cycles per instruction.
- If for each instruction type, we know its frequency (percentage) and number of cycles need to execute it, we can compute the overall CPI as follows:

$$CPI = \sum_{i=1}^{n} CPI_i \times F_i$$

- For example

Op	F_i	CPI_i	$CPI_i \times F_i$	% Time
ALU	50%	1	.5	23%
Load	20%	5	1.0	45%
Store	10%	3	.3	14%
Branch	20%	2	.4	18%
Total	100%		2.2	100%

Poor Performance Metrics

- Marketing metrics for computer performance included MIPS and MFLOPS
- MIPS : millions of instructions per second
  - MIPS = instruction count / (execution time x 10^6)
  - For example, a program that executes 3 million instructions in 2 seconds has a MIPS rating of 1.5
  - Advantage : Easy to understand and measure
  - Disadvantages : May not reflect actual performance, since simple instructions do better.
- MFLOPS : millions of floating point operations per second
  - MFLOPS = floating point operations / (execution time x 10^6)
  - For example, s program that executes 4 million instructions in 5 seconds has a MFLOPS rating of 0.8
  - Advantage : Easy to understand and measure
  - Disadvantages : Same as MIPS,only measures floating point

Mhz (MegaHertz) and Ghz (GigaHertz)

- 1 Hertz = 1 cycle per second
- 1 Ghz is 1 cycle per nanosecond, 1 Ghz = 1000 Mhz
  - (Micro-)architects often ignore dynamic instruction count…
- … but general public (mostly) also ignores CPI
  - Equates clock frequency with performance!
- Which processor would you buy?
  - Processor A: CPI = 2, clock = 5 GHz
  - Processor B: CPI = 1, clock = 3 GHz
  - Probably A, but B is faster (assuming same ISA/compiler)
- Classic example
  - 800 MHz PentiumIII faster than 1 GHz Pentium4!
  - More recent example: Core i7 faster clock-per-clock than Core 2
  - Same ISA and compiler!
- **Meta-point: danger of partial performance metrics!**

Benchmark Principles

- Desirable characteristics of a benchmark program:

1. It is written in a high-level language, making it portable across different machines
2. It is representative of a particular kind of programming domain or paradigm, such as systems programming, numerical programming, or commercial programming
3. It can be measured easily
4. It has wide distribution

System Performance Evaluation Corporation (SPEC)

- Benchmark suite
  - A collection of programs, defined in a high-level language
  - Together attempt to provide a representative test of a computer in a particular application or system programming area
- SPEC
  - An industry consortium
  - Defines and maintains the best known collection of benchmark suites aimed at evaluating computer systems
  - Performance measurements are widely used for comparison and research purposes

SPEC CPU2017

- Best known SPEC benchmark suite
- Industry standard suite for processor intensive applications
- Appropriate for measuring performance for applications that spend most of their time doing computation rather than I/O
- Consists of 20 integer benchmarks and 23 floating-point benchmarks written in C, C++, and Fortran
- For all of the integer benchmarks and most of the floating-point benchmarks, there are both rate and speed benchmark programs
- The suite contains over 11 million lines of code

Rate	Speed	Language	Kloc	Application Area
500.perlbench_r	600.perlbench_s	C	363	Perl interpreter
502.gcc_r	602.gcc_s	C	1304	GNU C compiler
505.mcf_r	605.mcf_s	C	3	Route planning
520.omnetpp_r	620.omnetpp_s	C++	134	Discrete event simulation - computer network
523.xalancbmk_r	623.xalancbmk_s	C++	520	XML to HTML conversion via XSLT
525.x264_r	625.x264_s	C	96	Video compression
531.deepsjeng_r	631.deepsjeng_s	C++	10	AI: alpha-beta tree search (chess)
541.leela_r	641.leela_s	C++	21	AI: Monte Carlo tree search (Go)
548.exchange2_r	648.exchange2_s	Fortran	1	AI: recursive solution generator (Sudoku)
557.xz_r	657.xz_s	C	33	General data compression

Table 2.5 (A)

SPEC CPU2017 Benchmarks

Kloc = line count (including comments/whitespace) for source files used in a build/1000 (Table can be found on page 61 in the textbook.)

Rate	Speed	Language	Kloc	Application Area
503.bwaves_r	603.bwaves_s	Fortran	1	Explosion modeling
507.cactuBSSN_r	607.cactuBSSN_s	C++, C, Fortran	257	Physics; relativity
508.namd_r		C++, C	8	Molecular dynamics
510.parest_r		C++	427	Biomedical imaging; optical tomography with finite elements
511.povray_r		C++	170	Ray tracing
519.ibm_r	619.ibm_s	C	1	Fluid dynamics
521.wrf_r	621.wrf_s	Fortran, C	991	Weather forecasting
526.blender_r		C++	1577	3D rendering and animation
527.cam4_r	627.cam4_s	Fortran, C	407	Atmosphere modeling
	628.pop2_s	Fortran, C	338	Wide-scale ocean modeling (climate level)
538.imagick_r	638.imagick_s	C	259	Image manipulation
544.nab_r	644.nab_s	C	24	Molecular dynamics
549.fotonik3d_r	649.fotonik3d_s	Fortran	14	Computational electromagnetics
554.roms_r	654.roms_s	Fortran	210	Regional ocean modeling.

Table 2.5 (B)

SPEC CPU2017 Benchmarks

Kloc = line count (including comments/whitespace) for source files used in a build/1000 (Table can be found on page 61 in the textbook.)

Benchmark	Base		Peak	
	Seconds	Rate	Seconds	Rate
500.perlbench_r	1141	1070	933	1310
502.gcc_r	1303	835	1276	852
505.mcf_r	1433	866	1378	901
520.omnetpp_r	1664	606	1634	617
523.xalancbmk_r	722	1120	713	1140
525.x264_r	655	2053	661	2030
531.deepsjeng_r	604	1460	597	1470
541.leela_r	892	1410	896	1420
548.exchange2_r	833	2420	770	2610
557.xz_r	870	953	863	961

Table 2.6

SPEC CPU 2017 Integer Benchmarks for HP Integrity Superdome X

(a) Rate Result (768 copies)

(Table can be found on page 64 in the textbook.)

Benchmark	Base		Peak	
	Seconds	Ratio	Seconds	Ratio
600.perlbench_s	358	4.96	295	6.01
602.gcc_s	546	7.29	535	7.45
605.mcf_s	866	5.45	700	6.75
620.omnetpp_s	276	5.90	247	6.61
623.xalancbmk_s	188	7.52	179	7.91
625.x264_s	283	6.23	271	6.51
631.deepsjeng_s	407	3.52	343	4.18
641.leela_s	469	3.63	439	3.88
648.exchange2_s	329	8.93	299	9.82
657.xz_s	2164	2.86	2119	2.92

Table 2.6

SPEC CPU 2017 Integer Benchmarks for HP Integrity Superdome X

(b) Speed Result (384 threads)

(Table can be found on page 64 in the textbook.)

Terms Used in SPEC Documentation

- Benchmark
  - A program written in a high-level language that can be compiled and executed on any computer that implements the compiler
- System under test
  - This is the system to be evaluated
- Reference machine
  - This is a system used by SPEC to establish a baseline performance for all benchmarks
    - Each benchmark is run and measured on this machine to establish a reference time for that benchmark
- Base metric
  - These are required for all reported results and have strict guidelines for compilation
- Peak metric
  - This enables users to attempt to optimize system performance by optimizing the compiler output
- Speed metric
  - This is simply a measurement of the time it takes to execute a compiled benchmark
    - Used for comparing the ability of a computer to complete single tasks
- Rate metric
  - This is a measurement of how many tasks a computer can accomplish in a certain amount of time
    - This is called a throughput, capacity, or rate measure
    - Allows the system under test to execute simultaneous tasks to take advantage of multiple processors

Figure 2.7

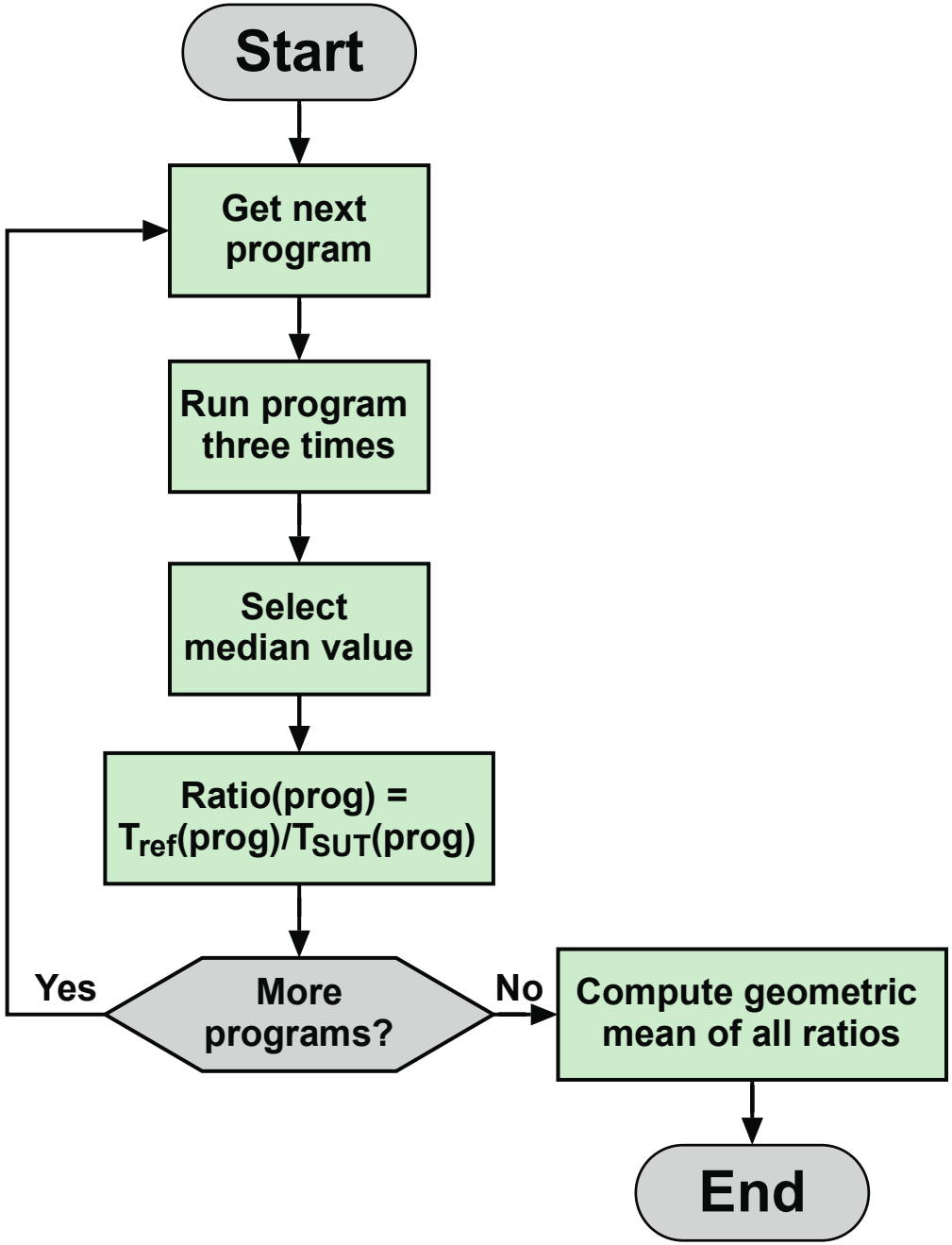


Figure 2.7 SPEC Evaluation Flowchart

Summary

Chapter 2

- Designing for performance
  - Microprocessor speed
  - Performance balance
  - Improvements in chip organization and architecture
- Multicore
- MICs
- GPGPUs
- Amdahl's Law
- Little's Law

Performance Concepts

- Basic measures of computer performance
  - Clock speed
  - Instruction execution rate
- Calculating the mean
  - Arithmetic mean
  - Harmonic mean
  - Geometric mean
- Benchmark principles
- SPEC benchmarks

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